

12312515 王洛源 第三次作业

Q1

(a) (i)

From :

$$z_1^{(i)} = W_1 x^{(i)} + b_1, \quad z_1^{(i)} \in \mathbb{R}^{D_{a_1} \times 1}$$

we have :

- $W_1 \in \mathbb{R}^{D_{a_1} \times D_x}$
- $b_1 \in \mathbb{R}^{D_{a_1} \times 1}$

From :

$$z_2^{(i)} = W_2 a_1^{(i)} + b_2, \quad z_2^{(i)} \in \mathbb{R}^{1 \times 1}$$

we have :

- $W_2 \in \mathbb{R}^{1 \times D_{a_1}}$
- $b_2 \in \mathbb{R}^{1 \times 1}$

(ii)

- $X \in \mathbb{R}^{D_x \times m}$ (each column is one example)
- $Y \in \mathbb{R}^{1 \times m}$
- $Z_1 \in \mathbb{R}^{D_{a_1} \times m}$
- $A_1 \in \mathbb{R}^{D_{a_1} \times m}$
- $Z_2 \in \mathbb{R}^{1 \times m}$
- $\hat{Y} \in \mathbb{R}^{1 \times m}$

(b)

Since $\hat{Y} = \sigma(Z_2)$, with $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, we have :

$$\frac{\partial J}{\partial Z_2} = \frac{\partial J}{\partial \hat{Y}} \odot \frac{\partial \hat{Y}}{\partial Z_2} = \frac{\partial J}{\partial \hat{Y}} \odot \hat{Y} \odot (1 - \hat{Y})$$

where $\odot$ denotes elementwise product.

Plugging in the given expression:

$$\frac{\partial J}{\partial Z_2} = -\frac{1}{m} \left(\frac{Y}{\hat{Y}} - \frac{1-Y}{1-\hat{Y}} \right) \odot \hat{Y} \odot (1-\hat{Y})$$

Elementwise, for each entry,

$$-\left(\frac{Y}{\hat{Y}} - \frac{1-Y}{1-\hat{Y}} \right) \hat{Y}(1-\hat{Y}) = -(Y(1-\hat{Y}) - (1-Y)\hat{Y}) = \hat{Y} - Y$$

Therefore, in matrix form,

$$\frac{\partial J}{\partial Z_2} = \frac{1}{m}(\hat{Y} - Y)$$

with shape $1 \times m$.

Now, $Z_2 = W_2 A_1 + b_2$, so each column is $z_2^{(i)} = W_2 a_1^{(i)} + b_2$. In matrix calculus, the gradient of J with respect to W_2 is

$$\frac{\partial J}{\partial W_2} = \frac{\partial J}{\partial Z_2} A_1^\top = \frac{1}{m}(\hat{Y} - Y) A_1^\top$$

where

- $\frac{\partial J}{\partial Z_2} \in \mathbb{R}^{1 \times m}$
- $A_1^\top \in \mathbb{R}^{m \times D_{a_1}}$

so $\frac{\partial J}{\partial W_2} \in \mathbb{R}^{1 \times D_{a_1}}$ matching the shape of W_2 .

Gradient descent update rule for W_2 with learning rate η :

$$W_2 \leftarrow W_2 - \eta \frac{\partial J}{\partial W_2} = W_2 - \eta \cdot \frac{1}{m}(\hat{Y} - Y) A_1^\top$$

Q2

(a)

Notice that

$$\{\forall h : |\varepsilon(h) - \hat{\varepsilon}(h)| \leq \gamma\}$$

is the complementary set of:

$$\{\exists h : |\varepsilon(h) - \hat{\varepsilon}(h)| > \gamma\}$$

Thus,

$$\Pr(\forall h : |\varepsilon(h) - \hat{\varepsilon}(h)| \leq \gamma) = 1 - \Pr(\exists h : |\varepsilon(h) - \hat{\varepsilon}(h)| > \gamma)$$

Use the bound $(\star)$, If $\text{RHS} \geq 1 - \delta$, then,

$$\Pr(\exists h : |\varepsilon(h) - \hat{\varepsilon}(h)| > \gamma) \leq \delta$$

Because of $(\star)$, a sufficient condition is:

$$2k \exp(-2\gamma^2 n) \leq \delta$$

and we can get:

$$\begin{aligned} 2k \exp(-2\gamma^2 n) \leq \delta &\iff \exp(-2\gamma^2 n) \leq \frac{\delta}{2k} \\ &\iff -2\gamma^2 n \leq \log \frac{\delta}{2k} \\ &\iff 2\gamma^2 n \geq \log \frac{2k}{\delta} \\ &\iff n \geq \frac{1}{2\gamma^2} \log \frac{2k}{\delta} \end{aligned}$$

Thus, a required condition is:

$$n \geq \frac{1}{2\gamma^2} \log \frac{2k}{\delta}$$

(b)

Since (a), when:

$$n \geq \frac{1}{2\gamma^2} \log \frac{2k}{\delta}$$

the probability of

$$|\varepsilon(h) - \hat{\varepsilon}(h)| \leq \gamma$$

is at least $1 - \delta$ for every $h \in H$

for $h = \hat{h}$

$$\varepsilon(\hat{h}) \leq \hat{\varepsilon}(\hat{h}) + \gamma$$

for $h = h^*$

$$\hat{\varepsilon}(h^*) \leq \varepsilon(h^*) + \gamma$$

Since $\hat{h}$ is the ERM:

$$\hat{\varepsilon}(\hat{h}) \leq \hat{\varepsilon}(h^*)$$

Then we can get:

$$\varepsilon(\hat{h}) \leq \hat{\varepsilon}(\hat{h}) + \gamma \leq \hat{\varepsilon}(h^*) + \gamma \leq \varepsilon(h^*) + 2\gamma$$

Let

$$\gamma = \sqrt{\frac{1}{2n} \log \frac{2k}{\delta}}$$

Then

$$\varepsilon(\hat{h}) \leq \varepsilon(h^*) + 2\sqrt{\frac{1}{2n} \log \frac{2k}{\delta}}$$

holds with a at least $1 - \delta$ probability